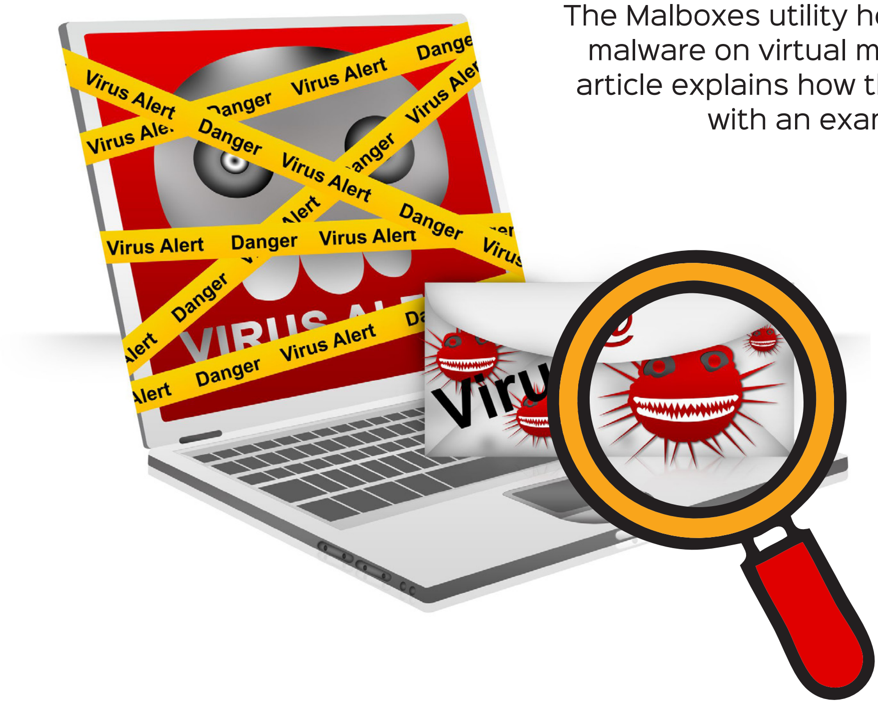
# Malboxes: Malware Analysis Made Easy




The Malboxes utility helps to analyse malware on virtual machines. This article explains how this tool works, with an example.

Imagine receiving good news like 'You have won a lottery' or 'You have been selected to receive an award' through an email. But the moment you open this harmless-looking mail, you realise all your files have been encrypted, and a ransom of over US$ 1,000 is being demanded to get them back. This is what happened to the victims of WanaCrypt0r, a global ransomware attack that first appeared on Friday, May 12, 2017.

It is very important to understand the nature of such malicious files. Malware analysis is similar to defusing bombs, as the goal is to deconstruct and decipher the software designed to cause harm to or spy on computer users. The program is frequently obfuscated (i.e., packed) to make analysis difficult, if not impossible.

Malware has the ability to do any or all of the following things.

- *Implant payload or variants:* In the field of malware, this refers to inserting a virus, worm, or Trojan designed to attack a victim's machine.
- *Obfuscate malware behaviour and code:* Malware obfuscation is a technique that makes textual and binary data harder to decipher. Attackers utilise such methods to make it difficult for the antivirus to detect it, and filter less malware.
- *Runtime polymorphism:* This is an encryption method used for mutating malware's static binary code in order to avoid discovery.
- *Spyware:* This collects information and data about the device and the user, while observing the user's activity, without the knowledge of the user.
- *Keylogging:* Every user activity including emails, accessed websites, apps and keystrokes is tracked.

## The need for malware analysis

Malware analysis is critical for investigating any occurrence of malware. It is carried out in two different ways — static and dynamic analysis.

***Static analysis:*** This is a simple way to dissect any binary file (executable) without executing it.

***Dynamic analysis:*** This involves execution of malware within the isolated environment (sandbox) that is separated from the real production environment.

Simple malware can easily be detected by using static analysis. So present day hackers use advanced techniques to obfuscate code and executables to bypass static detection techniques, making it imperative to use dynamic analysis.

If your organisation is attacked by malware, and the threat is eliminated by hiring a good security consultant, you still need to look out for the following to avoid another attack:

- Probability of rootkit or Trojan implantation in the system
- Is the threat eliminated?
- What changes did the attacker make to the system?
- How did the attacker access the backdoor to the system?